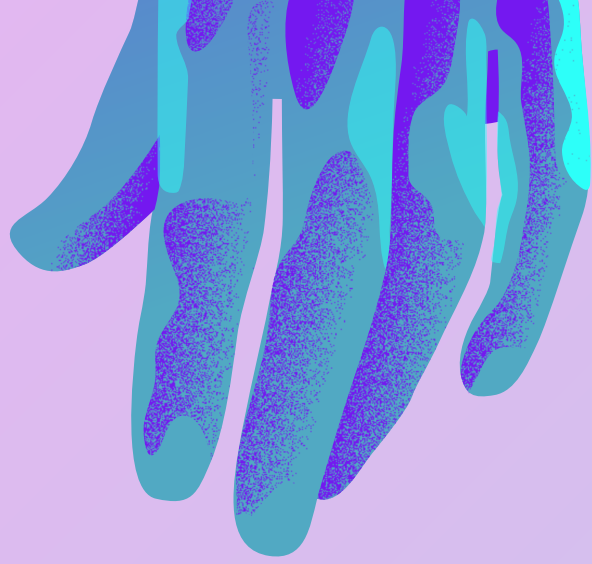




a multi-agent rag system for auditable token distribution

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research background and motivation

- Data inconsistency and delayed services (data silos among government departments)
- Lack of immutable audit trails affecting transparency in fund disbursement tracking
- Citizens' lack of control over personal information
- how ai agents can play a role in government services



news

Middle-class resisting Padu registration due to security and privacy concerns



NATION

Saturday, 17 Feb 2024

PETALING JAYA: There is still strong resistance from the M40 and T20 groups in registering on the Central Database Hub (Padu), with many citing security and privacy concerns, difficulty in registering, or, more worryingly, not seeing why they should bother.

Padu developed using outdated methods, says cybersecurity expert

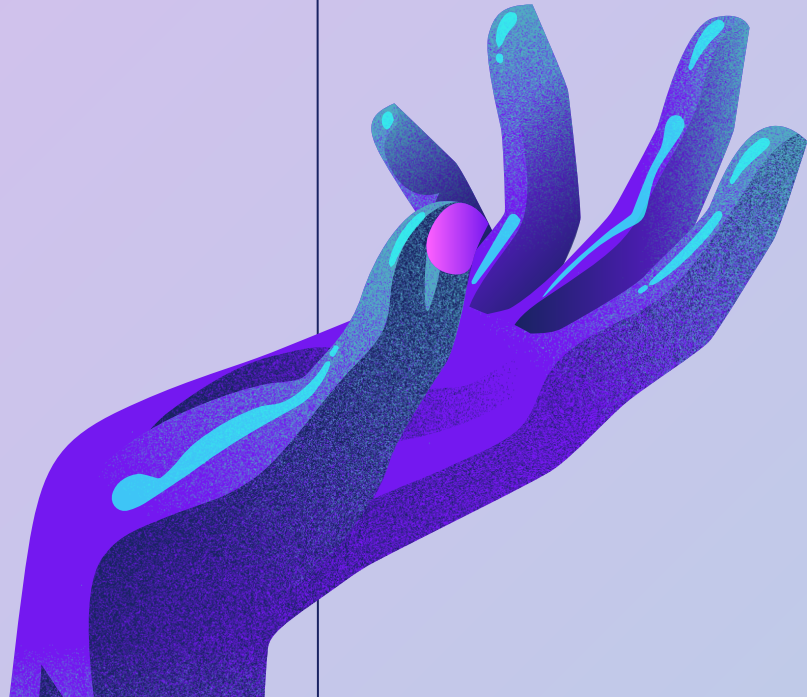
1 YEAR AGO

Chia Wan Rou



Share

Fong Choong Fook says Padu may be vulnerable due to outdated development methods, raising concerns about data integrity and security.



01.

To develop a **quantization method** to synthesize government policy **using ai agents + RAG and LLM** to determine eligibility for subsidies.

02.

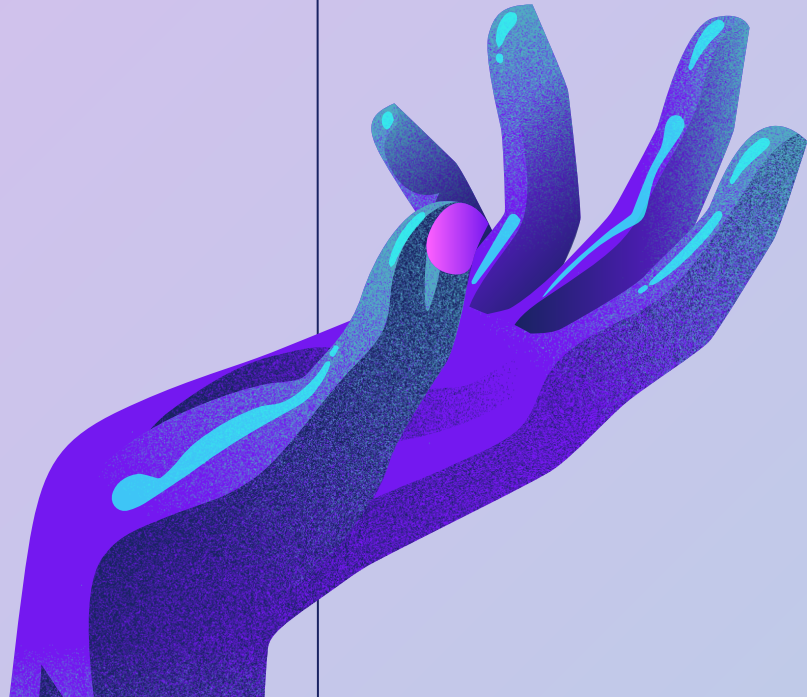
To **implement an ERC-20 token**, named MMYRC (Mock Malaysia Ringgit Coin), for airdropping funds to eligible citizens.

03.

To evaluate the application's effectiveness in **improving auditability and traceability of funds** compared to **existing systems**.



objectives of the
project



04.

to explore the use of **zero-knowledge proofs** (ZKPs) in a mock trusted setup with LHDN, allowing citizens to prove their income bracket (e.g., B40-B1, M40-M1, T20-T1) without revealing exact income values, thereby ensuring privacy preservation.

05.

To integrate **smolagents** (a agentic framework) into a dual-analysis framework that combines formula-based burden scoring with RAG-based reasoning, enabling comparison between interpretability and flexibility in eligibility assessments.

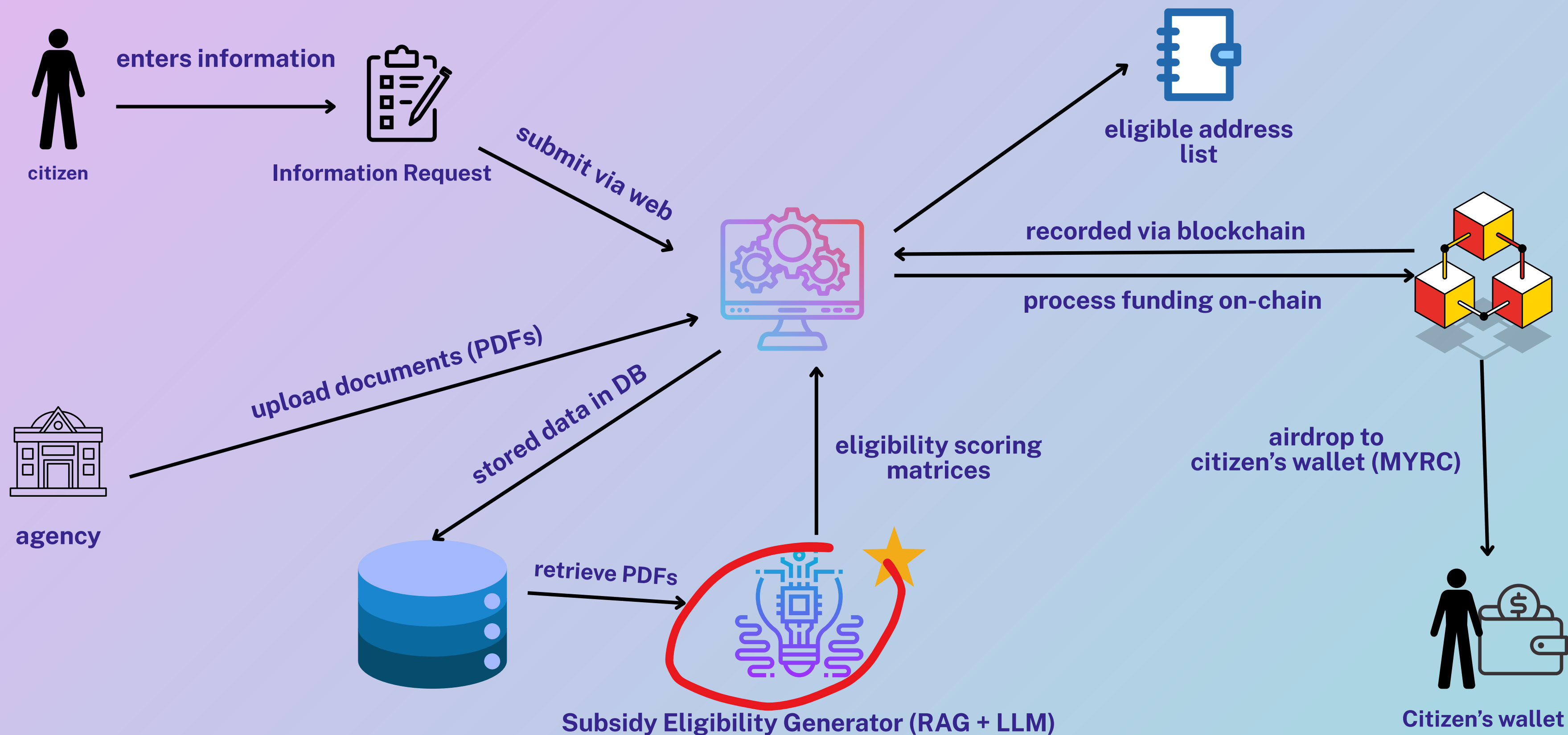


objectives of the
project

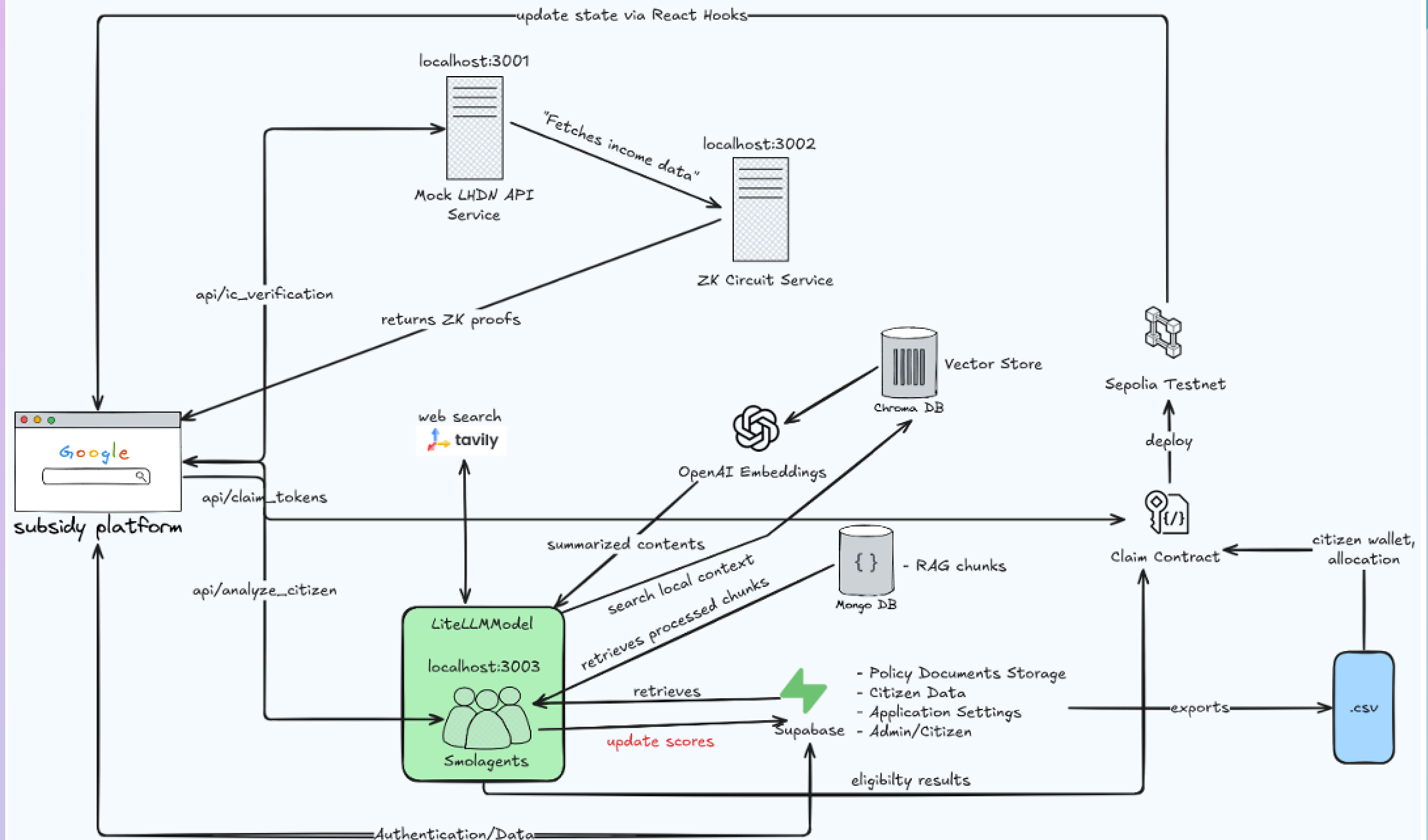
Feature	Traditional System (PADU)	Proposed Blockchain RAG System
Rule Setup	Manual, Static	Data-driven, LLM-generated
Transparency	Limited	Full on-chain traceability
Fraud Risk	High	Low(smart contract enforced)
Customisation per Region/Group	Hardcoded	Data-contextualized scoring
Real-time Auditability	No	Yes
Document Handling (PDFs)	Manual	Automated via RAG

Table 2.2 Comparison of proposed system over traditional PADU

proposed system architecture (fyp1)



new improvement (smolagents + zk) (fyp2)



zero-knowledge proofs (zkps): Balancing Privacy and Verification

Objective

To allow citizens to prove their income bracket (e.g., B40-B1, M40, T20) for subsidy eligibility without revealing the exact income value

The Purpose of the Mock LHDN API

1. IC Validation

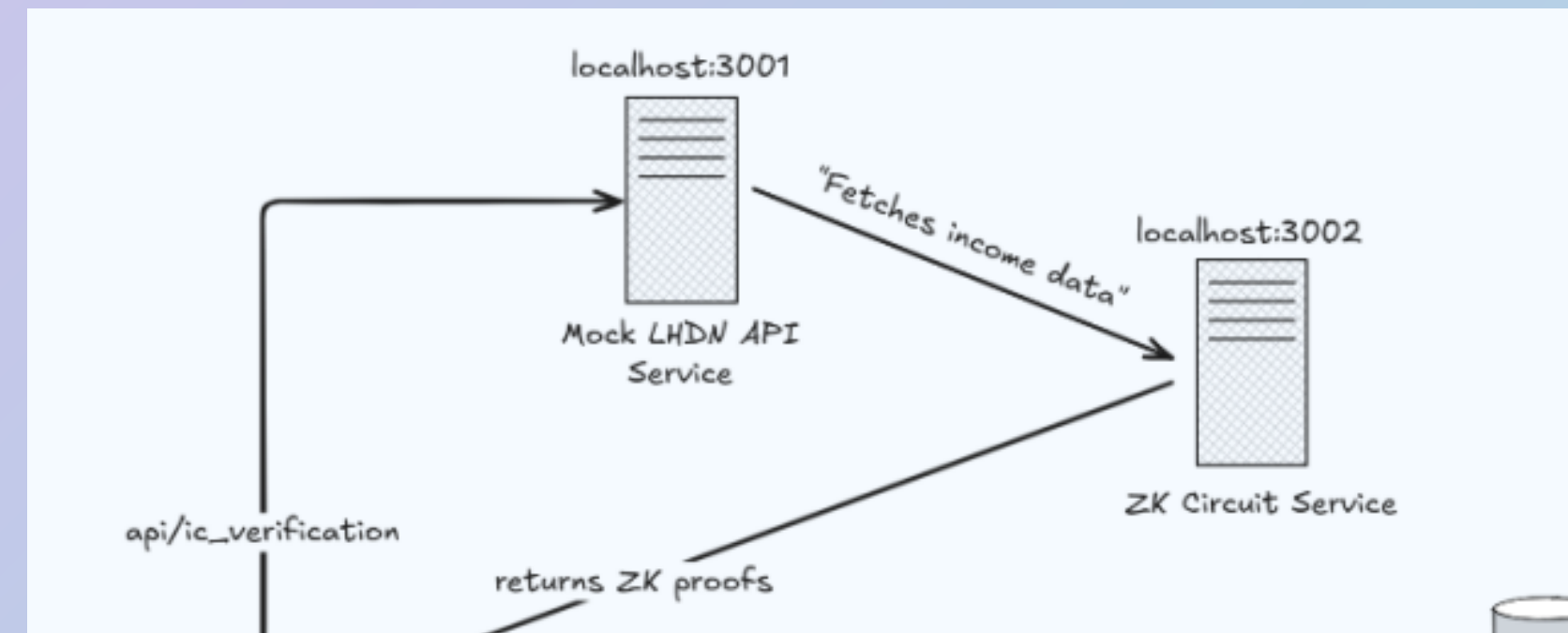
Performs rigorous format checking on the submitted Identity Card (IC) number.

2. Mock Income Lookup

Simulates government database functionality for development and testing environments, retrieving relevant financial information.

3. Data Security & Signing

Generates a digital signature (HMAC-SHA256) for the retrieved income data to ensure its integrity and authenticity.



System Components

Mock LHDN API Service

Port 3001

Simulated tax department integration

ZK Circuit Service

Port 3002

Zero-knowledge proof generation

ZKP Circuit Execution and Privacy Guarantee

ZKP Generation and Verification Workflow

Goal of ZK Circuit Service (Port 3002)

Generate cryptographic proof (using Groth16 zk-SNARKs) that demonstrates the citizen's income falls within a public bracket classification (B1-T2).

Input Preparation



Private Inputs (Hidden)

- Actual monthly_income (e.g., RM4,500)
- LHDN cryptographic signature



Public Inputs (Visible)

- IC hash identifier
- Malaysian income bracket thresholds (RM2,560, RM3,439, RM5,249...)

Core Circuit Logic (Witness Calculation)

1

Signature Verification Module

Ensures the LHDN signature is mathematically valid and checks data freshness

2

Income Threshold Comparison System

Uses LessThan components to evaluate private income vs public thresholds (e.g., $4500 < 5249 \rightarrow B4$ bracket)

3

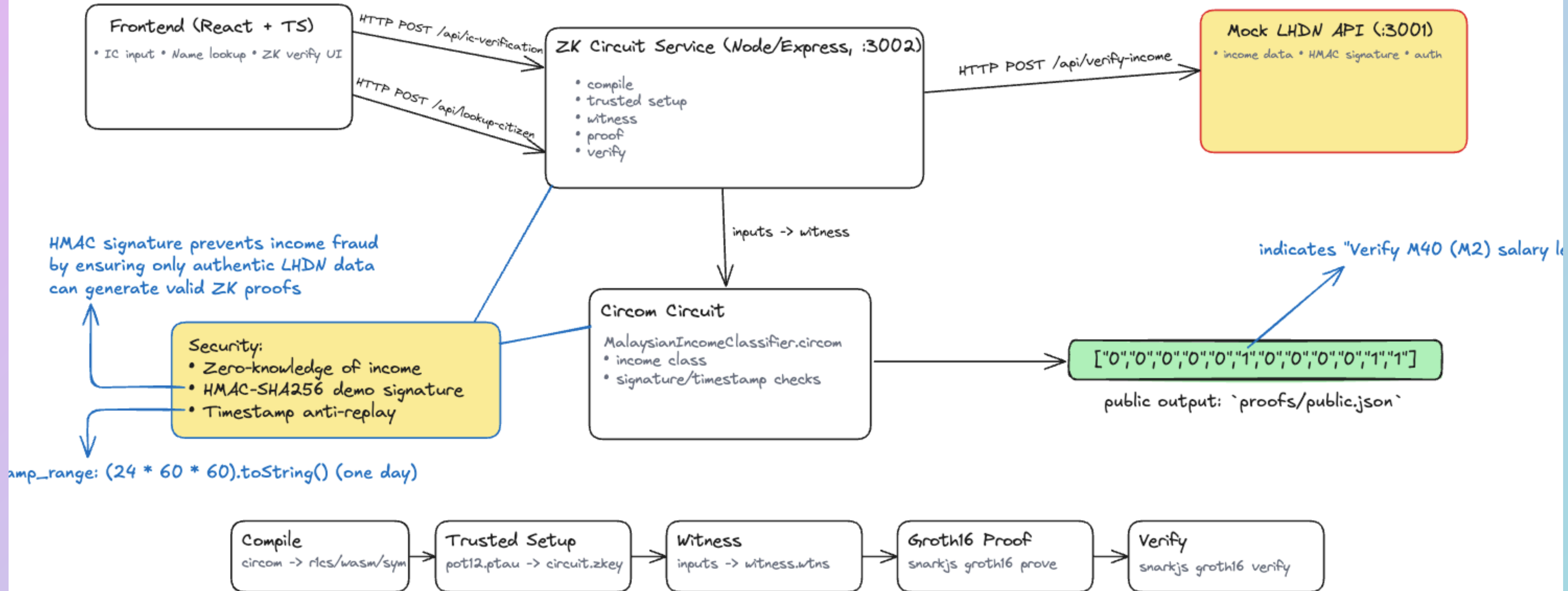
Security Gate

Conditional logic ensures classification output is valid only if LHDN signature verification succeeds

Proof Output and Verification

Groth16 Prover generates proof (Π) + public signal (income_bracket = B4). Verifier validates without accessing actual RM4,500.

ZK-SNARK Income Verification Architecture



INPUTS: Data Required to Generate the Proof

The circuit requires a mix of private (hidden) and public (visible) inputs to perform its cryptographic calculations and ensure data authenticity 7 8 .

1. Private Inputs (Hidden from Verifiers)

These inputs constitute the sensitive information that remains concealed throughout the verification process (the Zero-Knowledge property) 2

Input Parameter	Description	Source
monthly_income	The citizen's actual monthly income in Ringgit Malaysia (RM) 8 9 . 8	
signature	An HMAC signature generated by the Mock LHDN API to cryptographically authenticate the income data 8 . 8	
verification_timestamp	The timestamp indicating when the LHDN (Inland Revenue Board) verified and signed the income data 8 . 8	

2. Public Inputs (Visible to Verifiers)

These inputs are necessary for the verification process but do not compromise the citizen's privacy 8 10 .

Input Parameter	Description	Source
public_key	The LHDN's cryptographic public key used by the Signature Verification Module to check the signature authenticity 8 10 . 8 10	
ic_hash	A cryptographic hash of the citizen's Identity Card (IC) number used to bind the generated proof to a specific citizen identity 10 11 . 10	
timestamp_range	The maximum allowable age for the verification timestamp (e.g., 86400 seconds, representing one day) to prevent replay attacks and ensure data freshness 10 . 10	

OUTPUTS: Verification Signals and Classification Result

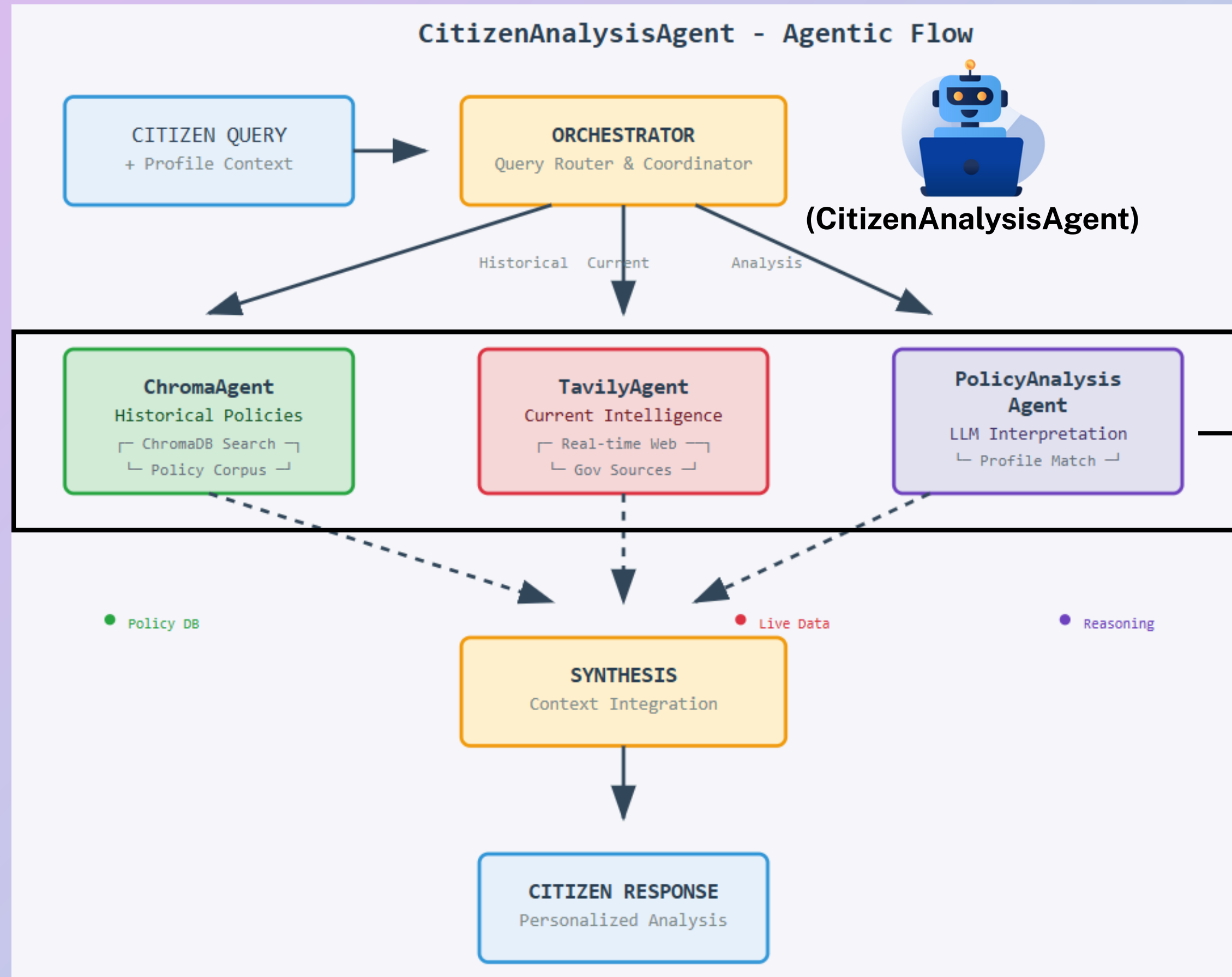
The circuit outputs signals that define the citizen's classification and confirm the integrity of the data used 12 13 .

Output Parameter	Description	Structure & Purpose	Source
class_flags 14	The core classification result, a 10-element array representing the one-hot encoded income bracket 12 13 .	Each position corresponds to brackets B1 through T2 (B1–B4, M1–M4, T1–T2). Exactly one bit is set to '1' to indicate the classification 12 15 . 12 13	
is_signature_valid	Boolean flag indicating whether the LHDN cryptographic signature is mathematically valid 13 .	Confirms the origin of the income data 13 . 13	
is_data_authentic	Boolean flag representing comprehensive data authenticity 13 .	Combines signature validity with timestamp and identity verification. If this flag is 0, all classification flags are zeroed out, preventing valid proofs from unverified data 12 16 . 13	

Example Output Interpretation 13 :

- If the output is ["0", "0", "0", "1", "0", "0", "0", "0", "0", "0", "1", "1"], it indicates:

agentic network for contextual policy reasoning (RAG Path)



AGENTIC WORKFLOW OUTPUTS

PHASE 3: Information Gathering

Historical Policies

Current Intelligence

Contextual Analysis



PHASE 4: Structured Output

Score

Numerical Rating

Classification

Category

Confidence

0.0 - 1.0

Reasoning

Explanation

Sources

Citations



PHASE 5: Comparison

Agentic Results vs Mathematical Formula



PHASE 6: Governance Insights

Edge Cases

Policy Nuances

Formula Gaps

implementation and testing results

 Sepolia Testnet Explorer

the project was developed as a working prototype, deployed on the Ethereum Sepolia Testnet using Visual Studio Code

successfully demonstrated the complete token distribution workflow. Total token supply capped at 1,000,000 MMYRC.



ZKP verification time is less than 1 second. The proof size is small (approx. 256 bytes) and remains constant regardless of input size, preserving computational efficiency

100,000 MMYRC were minted.
(10% of cap)

Eligible citizens were allocated 1,000 tokens each, demonstrating successful on-chain airdrops to verified eligible users



TRANSACTION ACTION

Transfer 100,000  MMYRC to  0xF79df53ae0d09b0219da032170Bf9F502d94009

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0xa0a7c10575607ec8cf428282e0315ff0753610ba5328d026cd75167f24fcc0db 

[illegible]

<https://sepolia.etherscan.io/tx/0x85b28bae7f2450d03c5b262c0406d6d3bc804cdea3ea6bffbefeb3d739818623#eventlog>

Transaction request

Estimated changes [?](#)

You receive **+1,000**  0x61B60...14F20

Request from [Alert >](#)

 HTTP localhost:5173

Interacting with [Alert >](#)

 0xeF79d...94009

Method [?](#)

Claim Tokens

Network fee [?](#)  0.0002 s SepoliaETH

\$0.83

Wallet Connected

0.010 ETH  0xCC...7dC6 [v](#)

Connected Address

0xCC06811c343Aa8F4CeB42c5d9053400C2Df27dC6

Current Balance

1000.00 MMYRC



Tokens Already Claimed

You have already claimed your MMYRC token allocation.



Tokens claimed successfully! Check your wallet balance.

impact and significance

transparency & auditability

Every transaction is immutably recorded on a public ledger, minimizing fraud risk and strengthening public trust in government processes

governance model

The project established a model for auditable, efficient, and citizen-centric public fund management by combining blockchain, AI-driven reasoning, and privacy-preserving technologies

citizen-centric design

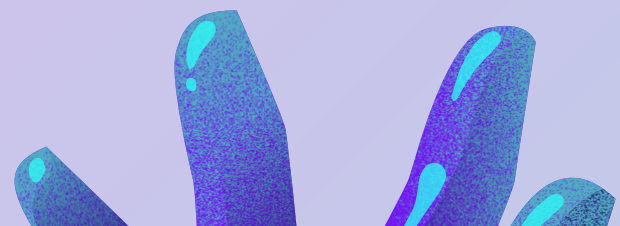
The system balances transparency with individual privacy through the use of ZKPs, allowing citizens to verify eligibility without disclosing sensitive financial data

contribution

- **Demonstrates feasibility of blockchain + AI subsidy distribution**
- **Lays foundation for future digital governance initiatives**

conclusion

The project successfully demonstrated the transformative potential of integrating blockchain and agentic frameworks into PADU to modernize subsidy distribution, achieving enhanced data integrity, transparency, and resilience.



recommendation

pilot deployment

Stage rollout across selected agencies

system integration

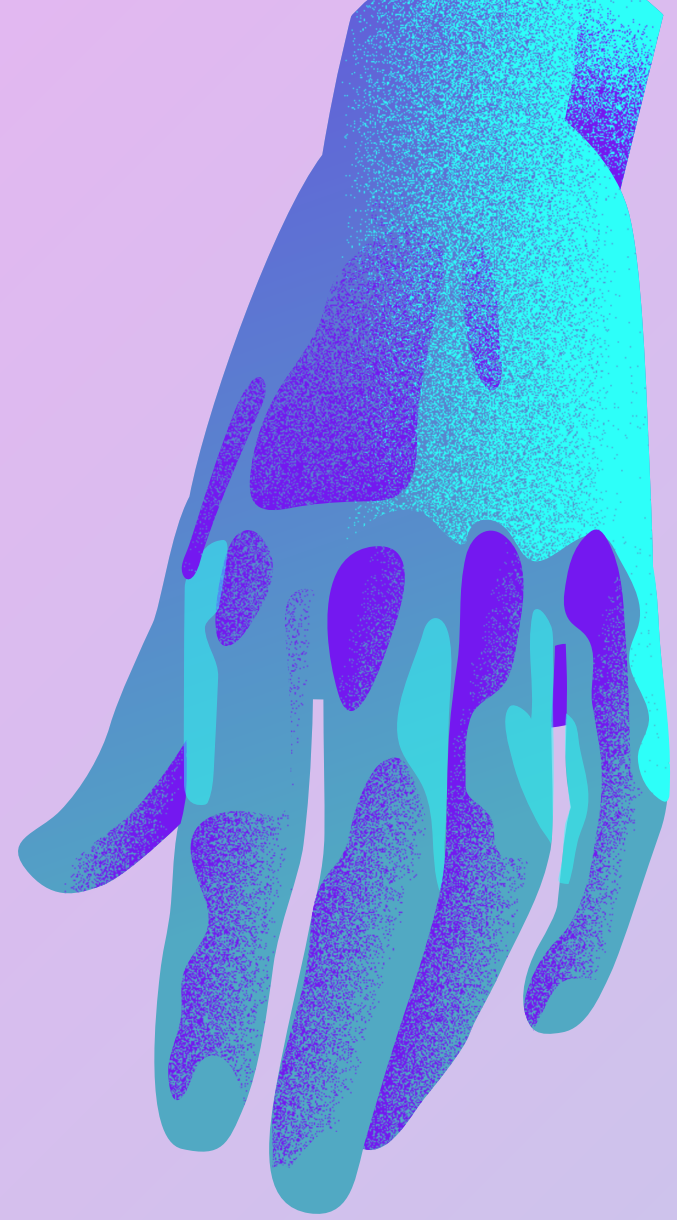
Link with national databases for seamless verification

security & trust

Perform smart contract audits
Explore broader use of ZKPs for citizen attributes

scalability & performance

Use Layer-2 solutions like zk-rollups
Optimize smart contracts & storage methods



**thank you
very much!**

